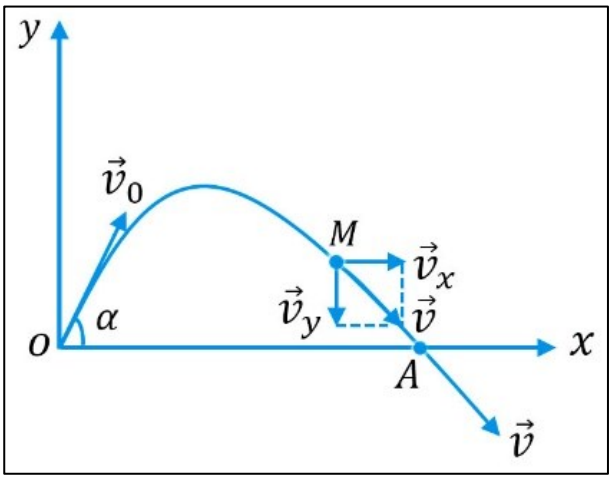
Physics IGCSE	Summary formula	Physics grade 10
Content	**Symbols**	**System international unit (SI unit)**
*** Projectile motion** **1.** Velocity-component of projectile motion.	1. $\begin{cases} v_x = v_{ox} = v_o \cos\alpha \\ v_y = v_{oy} - gt = v_0 \sin\alpha - gt \end{cases}$	Where, v_o is the initial velocity measured in (m/s). v_x is the ox-velocity measured in (m/s). v_y is the oy-velocity measured in (m/s). y is the height of projectile motion measured in (m). α is the angle of projectile motion measured in $(^\circ)$. g is the acceleration due to gravity measured in (m/s^2) x is the range of projectile motion measured in (m). x_m is the maximum range of projectile motion. y_m is the maximum height attained by projectile motion. t is the time taken measured in (s). v is the final velocity measured in (m/s). v_o is the initial velocity measured in (m/s).
2. Displacement in ox and oy axis after t second.	2. $\begin{cases} x = v_{ox}t = v_o \cos\alpha t \\ y = v_o \sin\alpha t - \frac{1}{2}gt^2 \end{cases}$	
3. Equation of path followed by particle in projectile motion.	3. $\begin{cases} y = v_o \sin\alpha t - \frac{1}{2}gt^2 \\ y = x\tan\alpha - \frac{gx^2}{2v_o^2\cos^2\alpha} \end{cases}$	
4. Time to reach maximum height and total time of flight.	4. $\begin{cases} t = \frac{v_o \sin\alpha}{g} \\ t = \frac{2v\sin\alpha}{g} \therefore \text{Tot-time flight} \end{cases}$	
5. Rage of projectile motion	5. $\begin{cases} x = \frac{v_o^2 \sin 2\alpha}{g} \\ x_{\max} = \frac{v_o^2}{g} \therefore \sin 2\alpha = \sin\frac{\pi}{2} \end{cases}$	
6. Maximum height attained	6. $\begin{cases} y_m = \frac{(v_o\sin\alpha)^2}{2g} \\ y_m = \frac{v_o^2}{2g} \end{cases}$	
7. Instantaneous velocity	7. $\begin{cases} v^2 - v_o^2 = +2gy \ (\vec{g} \uparrow\uparrow \vec{oy}) \\ v^2 - v_o^2 = -2gy \ (\vec{g} \uparrow\downarrow \vec{oy}) \end{cases}$	
*** Cricle Motion** **8.** Centripetal acceleration	8. $\begin{cases} a_c = \frac{v^2}{r} \\ a_c = \omega v = \frac{2\pi v}{T} \\ a_c = \omega^2 r = \frac{4\pi^2 r}{T^2} \end{cases}$	a_c is the centripetal acceleration measured in (m/s^2). $F = F_{net}$ is the net force measured in (N). v is the velocity of an object measured in (m/s). r is the radius of a circle measured in (m). m is the mass of an object measured in (kg). T is the period complete one rev measured in (s).
9. Second's law of Newton	9. $\begin{cases} F_{net} = ma_c \\ F_{net} = m\frac{v^2}{r} \end{cases}$	
10. Relative velocity	10. $v_{a/b} + v_{b/c} = v_{a/c}$	

*** Electrostatic**

11. Ohm's law

$$11.\begin{cases} V = RI \\ I = \dfrac{V}{R} \\ R = \dfrac{V}{I} \end{cases}$$

12. Electrical power

$$12.\begin{cases} P = VI \\ P = RI^2 \\ P = \dfrac{V^2}{R} \end{cases}$$

13. Electrical energy

$$13.\begin{cases} E = VIt \\ E = RI^2t \\ E = \dfrac{V^2}{R}t \end{cases}$$

14. Cost of electricity bills

$$14.\begin{cases} \text{cost=E(kWh)}\times\text{price/1kWh} \\ 1kWh = 3600000\,J \end{cases}$$

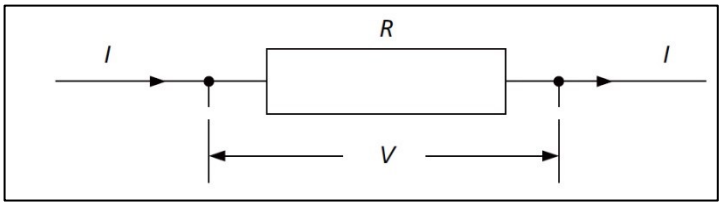
15. Series circuit

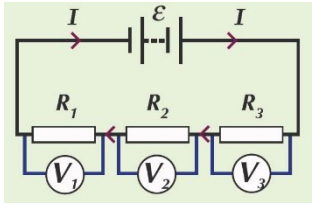
$$15.\begin{cases} I = I_1 = I_2 = I_3 = \ldots = I_n \\ V = V_1 + V_2 + V_3 + \ldots + V_n \\ R = R_1 + R_2 + R_3 + \ldots + R_n \end{cases}$$

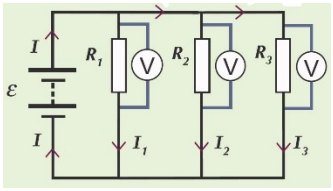
16. Parallel circuit

$$16.\begin{cases} I = I_1 + I_2 + I_3 + \ldots + I_n \\ V = V_1 = V_2 = V_3 = \ldots = V_n \\ \dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3} + \ldots + \dfrac{1}{R_n} \end{cases}$$

V is the voltage measured in (V).
R is resistance measured in (Ω).
I is the electric current measured in (A).
P is the electrical power measured in (W).
E is electrical energy measured in (J).
t is the time taken measured in (s).



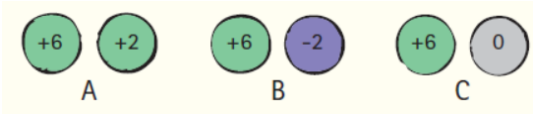




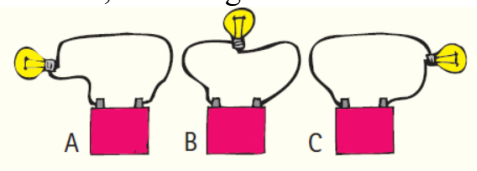
REVIEW FOR FINAL EXAM GRADE 10

A. Multiple choice questions

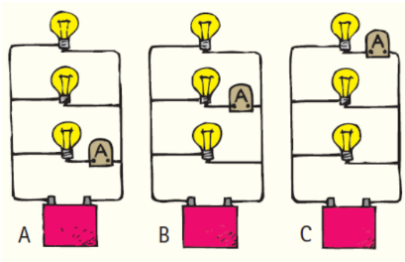
1. The three pairs of metal, same-size spheres have different charges on their surfaces as indicated. Each pair is brought together, allowed to touch, and then separated. Rank from greatest to least the total amount of charge on the pairs of spheres after separation.



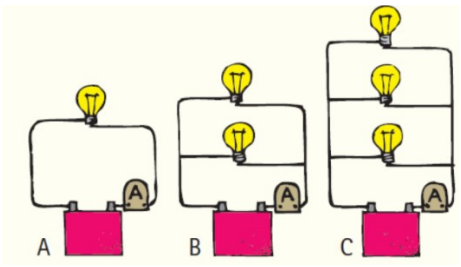
2. Rank the circuits according to the brightness of the identical bulbs, from brightest to dimmest.



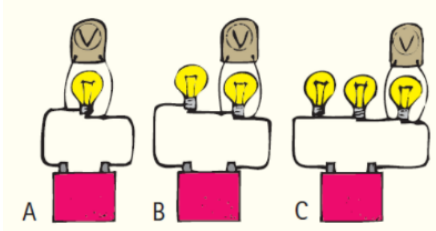
3. All the bulbs in the three circuits are identical. An ammeter is placed in different locations, as shown. Rank the current readings in the ammeter from highest to lowest.



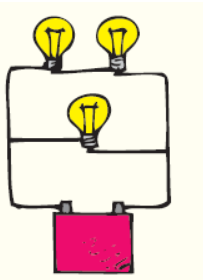
4. All bulbs are identical in the three circuits. An ammeter is connected next to each battery as shown. Rank the current readings in the ammeter from highest to lowest.



5. All bulbs are identical in the circuits shown. A voltmeter is connected across a single bulb to measure the voltage drop across it. Rank the voltage readings from highest to lowest.

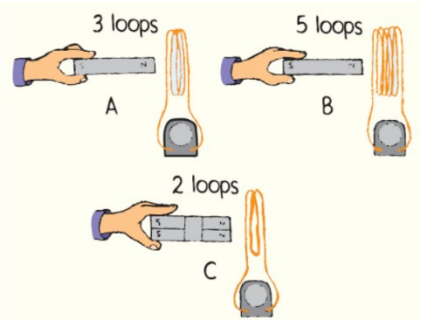


6. Consider the three parts of the circuit: (1) the top branch with two bulbs, (2) the lower branch with one bulb, and (3) the battery.

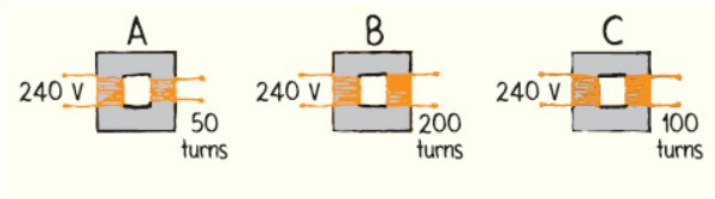


(a) Rank the current through each part, from greatest to least.
(b) Rank the voltage across each, from greatest to least.

7. Bar magnets are moved into the wire coils in identical quick fashion. Voltage induced in each coil causes a current, as indicated on the galvanometer. Neglect the electrical resistance in the loops in the coil, and rank from highest to lowest the readings on the galvanometer.



8. The three transformers are each powered with 100 W, and all have 100 turns on the primary. The number of turns on each secondary varies as indicated.
(a) Rank the voltage output of the secondaries from greatest to least.
(b) Rank the current in the secondaries from greatest to least.
(c) Rank the power output in the secondaries from greatest to least.



9. The source of all magnetism is
(a) tiny bits of iron. (b) tiny domains of aligned atoms.
(c) small lodestones. (d) the motion of electrons.

10. Surrounding moving electric charges are
(a) electric fields. (b) magnetic fields.
(c) both of these (d) neither of these

11. The essential physics concept in an electric generator is
(a) Coulomb's law. (b) Ohm's law.
(c) Faraday's law. (d) Newton's second law.

12. A vector in an x-y coordinate can be split into …
a. one component b. two components
c. two or three components (depends on the angle of the vector)

13. Current in a circuit if resistance of 15 Ω and potential difference of 30.0 V is applied across its ends is
a. 1 A b. 2 A c. 0.5 A d. 0.20 A

14. A 1000kg car enters an 80m radius curve at 20m/s. What is the centripetal force must be Supplied by the friction so the car does not skid?
a. 5N b. 250N c. 5000N d. 1000N

15. The Current in a 1000 W-10Ω light bulb power supply is:

a. 10A b. 100 A c. 1000A d. 0 A

16. The current is ……in the electric circuit.

a. Flow of electric Charges b. Coulombs

c. Complete circuit d. amount of electric charges

17. Which of the following is true of electrical forces?
a. Like charges attract, unlike charges repel.

b. Electrical forces are produced by electrical charges.

c. Positive and negative charges can combine to produce the third type

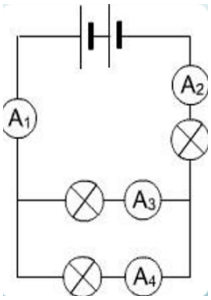
d. Electrical forces are weaker than gravitational forces.

18. Direction of conventional current is
a. direction of neutron flow
b. direction of electron flow
c. direction of flow of positive charge
d. same as that of electric current

19. The voltage change from positive to negative and back again with a ……of 50Hz.
a. Electron changes direction b. Cable
c. Frequency d. Alternating current

20. An uncharged object has
a. more protons b. more electrons
c. equal electrons and protons
d. no protons and electrons

21. What would ammeter 1 read if the current at ammeter 3 was 0.4A and the current at ammeter 4 was 0.4A?
a. 0.4A b. 0A c. 0.8A
d. None of the above



22. A 50.0 Ω resistance causes a current of 5.00 mA to flow through a circuit connected to a battery. What is the power in the circuit?
a. $1.00\times10^{-2}W$ b. $1.25\times10^{-3}W$
c. $1.00\times10^{-3}W$ d. $2.50\times10^{-3}W$

23. A 100-W lightbulb is connected to a 120-V electric line. What is the current that the lightbulb draws?
a. $0.8\,A$ b. $1.2A$
c. $1.0\,A$ c. $2A$

24. A 5.0 Ω resistor is connected to a 9.0-V battery. How much thermal energy is produced in 7.5 min?
a. $1.2\times10^{2}J$ b. $3.0\times10^{3}J$
c. $1.3\times10^{3}J$ d. $7.3\times10^{3}J$

25. How much electric energy is delivered to a 60.0-W lightbulb if the bulb is left on for 2.5 hours?
a. $4.2\times10^{-2}J$ b. $1.5\times10^{2}J$
c. $2.4\times10^{1}J$ d. $5.4\times10^{5}J$

26. A magnetic force acts most strongly on a current-carrying wire when the wire
(a) carries a very large current.
(b) is perpendicular to the magnetic field.
(c) either or both of these
(d) none of the above

27. When a pair of charged particles are brought twice as close to each other, the force between them becomes
(a) twice as strong. (b) four times as strong.
(c) half as strong. (d) one quarter as strong.

28. An electric field surrounds all
(a) electric charge. (b) electrons.
(c) protons. (d) all of these

29. If you double both the current and the voltage in a circuit, the power
(a) remains unchanged if the resistance remains constant.
(b) halves.
(c) doubles.
(d) quadruples.

30. A 1.60-m-tall girl throws a football at an angle of 41.0° from the horizontal and at an initial velocity of 9.40 m/s. How far away from the girl will it land?

a. $4.55\,m$ b. $8.90\,m$

c. $5.90\,m$ c. $10.5\,m$

31. A dragonfly is sitting on a merry-go-round 2.8 m from the center. If the tangential velocity of the ride is 0.89 m/s, what is the centripetal acceleration of the dragonfly?

a. $0.11\,m/s^2$ b. $0.32\,m/s^2$

c. $0.28\ m/s^2$ c. $2.2\ m/s^2$

32. The centripetal force on a 0.82-kg object on the end of a 2.0-m massless string being swung in a horizontal circle is 4.0 N. What is the tangential velocity of the object?

a. $2.8\,m/s^2$ b. $4.9\,m/s^2$

c. $3.1\,m/s^2$ c. $9.8\,m/s^2$

33. A 1000-kg car enters an 80.0-m-radius curve at 20.0 m/s. What centripetal force must be supplied by friction so the car does not skid?

a. $5.0\,N$ b. $5.0\times10^3\,N$

c. $2.5\times10^2\,N$ c. $1.0\times10^3\,N$

B. Critical thinking questions.

34. If the two feet of a bird on the high-potential wire of a power line are widely spaced, won't it get a shock?

35. What is the net charge of a charged capacitor?

36. What quantity is measured in kilowatt-hours? What quantity is measured in horse-power(hp)?

37. Why do clothes often cling together after tumbling in a clothes dryer?

38. Draw a small bar magnet and show the magnetic field lines as they appear around the magnet. Use arrows to show the direction of the field lines.

C. Work problems

39. The element of 500W electric iron is designed for use on a 100 V supply. What value of another resistance is needed to be connected in series with electric iron in order that the iron can be operated from 240 V supply?

40. The following are the details of load on a circuit connected through a supply meter:

(i) Six lamps of 50W each working for 4 hours per day

(ii) Two fluorescent tubes 100W each working for 4 hours per day

(iii) One 1000W heater working for 3 hours per day If each unit(1kWh) of energy costs 700Riels, what will be the electricity bill for the month of June?

41. Calculate the cost of operating a 12.5kW machine continuously for a period of 8.5hours and a 1000W motor for a period of 2hours if the cost per unit is 700 Riels.

42. A ball of mass 0.5 kg is tied to a thread and rotated at a constant angular velocity of 10 rad/s in a circle of radius 1 m. Use this data to determine the answers to questions 10 and 11

The centripetal acceleration is:

a. $50\,m/s^2$ b. $100/2\pi\ m/s^2$

c. $50/2\pi\ m/s^2$ c. $100\ m/s^2$

The tension in the thread is:

a. $25\,N$ b. $50/2\pi\ N$

c. $25/2\pi\ N$ c. $50\,N$

43. A current of 1.4 A when flowing through a circuit for 15minutes dissipates 200 kJ of energy, calculate:

a. The voltage

b. The power dissipated

c. The resistance of the circuit

44. An object is suspended by a thread 250 mm long and both object and thread move in a horizontal circle with a constant angular velocity of 2.0 rad/s. If the tension in the thread is 12.5 N, determine the mass of the object.

45. A vehicle of mass 750 kg travels round a bend of radius 150 m, at 50.4 km/h. Determine the centripetal force acting on the vehicle.

46. An aircraft is turning at constant altitude, the turn following the arc of a circle of radius 1.5 km. If the maximum allowable acceleration of the aircraft is 2.5 g, determine the maximum speed of the turn in km/h. Take g as 9.8 m/s.

47. An object is suspended by a thread 400 mm long and both object and thread move in a horizontal circle with a constant angular velocity of 3.0 rad/s. If the tension in the thread is 36 N, determine the mass of the object.

48. Calculate the centripetal force acting on a vehicle of mass 1 tonne when travelling round a bend of radius 125 m at 40 km/h. If this force should not exceed 750 N, determine the reduction in speed of the vehicle to meet this requirement.

49. A speed-boat negotiates an S-bend consisting of two circular arcs of radii 100 m and 150 m. If the speed of the boat is constant at 34 km/h, determine the change in acceleration when leaving one arc and entering the other.